Vishnu AP
Data Scientist

Contact

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Skills

- Computer Vision
- NLP
- Predictive Modeling
- Statistics
- Data Visualization
- Big Data

Summary

- Qualified and result oriented professional with an experience of **10+ years** in Data Science/Data Analytics projects.
- Very Strong Technical Skills in developing data driven solutions for Business Problem solving.
- Core competencies include **Under water Image Processing, Computer Vision, Predictive Modeling, Statistical Analysis, Data Visualizations, Big Data ...etc**.
- **Certified in Data Science from NIT Kurukshetra** (Link)
- **Microsoft Certified: Azure Data Scientist Associate** (Link)
- **Data Bricks Certification: Generative AI Fundamentals** (Link)
- Corporate training experience for **6+ Years**

Skills

Domain	**Finance, Insurance, Health Care, IT Operations, Mari culture**
Programming / Scripting	**Python, R, Java Script, SQL, QlikSense, Go**
Databases	**SQL Server, MySQL, Oracle, MariaDB, MongoDB, Elastic Search, Bigquery**
Visualization / Reporting	**QlikView, QlikSense, PoweBI, ELK Stack, Nprinting, Grafana**
Cloud	**Microsoft Azure, GCP**
Others	**Data Analysis, Statistics, Hypothesis Testing, Machine Learning, Computer Vision, Object detection, Keypoint Detections, NLP, Gen AI, LLM, BigqueryML, GitHub, CICD Pipelines, Docker, Kubernates, MLOps, Vertex AI, AutoML, PySpark, PysparkML, Scikit-learn, Tensorflow, Pytorch, Flask, Fast API,**

Education

Course	Collage/University	CGPA/Marks
MTech (Integrated)	Vellore Institute of Technology	8.29
BCA	St. Claret College	80%

Languages

	English	Hindi	Malayalam	Tamil	Kannada
Speak	✓	✓	✓	✓	✓
Read-Write	✓	✗	✓	✗	✗

Work History

Data Scientist - Predikly Technologies (2021 – 2025)

Specialized in AI-driven solutions for the **Mari culture industry**, supporting clients farming species like Salmon, Sea Bream, and Yellow Tail.

Core responsibilities:

- Developing state-of-the-art detection models for fish density, key points, biomass, sea lice, disease, welfare, and pellets...etc.
- Performing bounding box tightness analysis and annotation comparison to ensure annotation quality and data integrity.
- Maintaining Customized CVAT for optimized annotation workflows tailored to aquaculture needs and Configuration and maintenance of Auto Annotation Models.
- Designing and implementing data pipelines and interactive dashboards to support Quality, Performance, SLA compliance and actionable insights.

Senior Software Engineer - Nous Info Systems (2019 – 2021)

Providing analytical and AI-driven solution services to clients across diverse domains, including **Finance, Healthcare, Insurance, and E-commerce**. This role involves leveraging data to drive insights, optimize operations, and enable data-driven decision-making.

Core Responsibilities:

- Designing and implementing ETL pipelines to extract and integrate data from various sources.
- Utilizing web scraping techniques to gather publicly available data for analysis.
- Conducting data modeling and analysis to uncover trends and actionable insights.
- Developing, fine-tuning, and maintaining machine learning models to address specific business needs.
- Creating interactive visualizations and automating reports to deliver better business insights and decision-making tools.

Consultant Technical - Wipro Technologies (2014 – 2019)

Delivering analytical and AI-driven solutions to clients across diverse domains, including **IT Operations, Finance, and Resource Management.** This role focuses on transforming client requirements into actionable insights and enabling data-driven decision-making.

Core Responsibilities:

- Understanding and consolidating client requirements to derive key performance indicators (KPIs) for effective monitoring and evaluation.
- Developing interactive dashboards for IT Operations, providing insights from tools such as monitoring systems, event management platforms, ticketing systems, backup solutions, and antivirus tools.
- Building predictive and statistical models to automate processes and reduce manual effort, enhancing operational efficiency.
- Conceptualizing, developing, testing, and deploying reports that equip top and mid-level management with critical business insights and strategic overviews.

Projects & POCS

Fish Density Analysis

Delivering computer Vison solution to analyze biomass density by detecting the salmon head from a specific camera angle set by the configured petrol path during the feeding process.

Technologies: OpenCV, Python, TensorFlow, CVAT, GCP Resources.

Model: RetinaNet (Restnet Backbone)

Area: Object Detection, Under water Image Processing

Impact: Analyzing the biomass density during the feeding process at the feeding area helps to optimize the feeding amount. This is one of the metrics basically influences feeding rate parameter (FR) with rules like reduce the feeding rate (FR) in a low density scenario. This basically helps to save the wastage of the feed.

Feed Fall through Detection

Delivering computer Vison solution to analyze pellet fall through by detecting the pellet from a specific camera angle set by the configured petrol path during the feeding process.

Technologies: OpenCV, Python, TensorFlow, CVAT, GCP Resources.
Model: RetinaNet (Spinenet Backbone)
Area: Object Detection, Under water Image Processing
Impact: Analyzing the pellet fall through during the feeding process below the feeding area helps to optimize the feeding amount. This is one of the metrics basically influences feeding rate parameter (FR) with rules like reduce the feeding rate (FR) in a high fall through. This basically helps to save the wastage of the feed.

Welfare Analysis Model

Delivering computer Vison solution to analyze the condition of the fishes by observing certain scenarios like presence wound, scale loss, dorsal fin damage, vertebral deformity

Technologies: OpenCV, Python, TensorFlow, CVAT, GCP Resources.
Model: RetinaNet (Spinenet Backbone)
Area: Object Detection, Under water Image Processing
Impact: Analyzing these measures on fish helps to maintain the quality of fishes desired by the market and helps to take necessary actions to avoid these kind of event occurrences

Disease Analysis Model

Delivering computer Vison solution to analyze the condition of the fishes by observing certain scenarios like presence lump sucker, lice matured, lice young, lice egg strings
Technologies: OpenCV, Python, TensorFlow, CVAT, GCP Resources.
Model: RetinaNet (Spinenet Backbone)
Area: Object Detection, Under water Image Processing
Impact: Analyzing these measures on fish helps to maintain the quality of fishes desired by the market and helps to take necessary actions to avoid further spread of these infections.

Biomass Analysis with Keypoints

Delivering computer Vison solution to the calculation of biomass of the fishes by identifying certain key points like lip, eye, dorsal fin, upper caudal peduncle, center caudal fin, lower caudal peduncle, anal fin, pelvic fin, pectoral fin. Based on the key points captured using stereo camera and calculated depth the biomass is being determined

Technologies: OpenCV, Python, TensorFlow, CVAT, GCP Resources.
Model: Custom RetinaNet (Spinenet Backbone)
Area: Object Detection, Under water Image Processing
Impact: Based on the biomass calculation we determine the amount of feed to be given, Feed Growth Ratio, Harvesting Period Estimation etc.

Insurance QA Chatbot

AI powered virtual assistant designed to customer queries related to insurance policies, claims, coverage and procedures. Leverages Large Language Models (LLM) and Retrieval Augmented Generation (RAG) to provide accurate, real-time responses by retrieving relevant information's from knowledge base.

Technologies: Python, LangChain, Faais
Model: Mistral 7B
Area: Predictive Modeling - Classification
Impact: Human resource reduction on the customer support. The customers can enjoy quick and accurate information within minimum response time.

ITOP Ticket Routing System

An intelligent ticket routing system analyze the description of the raised ticket by the users and route the ticket to the right team for resolution.

Technologies: Python
Model: Custom LSTM
Area: Text Classification
Impact: Initially this routing was being handle by manually by Service Desk. With the new model the Service Desk operations could be reduced, avoiding human errors from the system, Reduction in ticket hoping and Resolution in minimum time.